

SPATIO-TEMPORAL CHANGE ANALYSIS OF WETLAND IN CDA BOUNDARY BY REMOTE SENSING TECHNIQUE

Background of the Study

A wetland is an area of land that is saturated with water either permanently or seasonally. Wetlands can be freshwater, brackish (partly salty), or saline (very salty). Chattogram City has been suffering from many environmental problems including flooding, water logging and other related problems. A few decades ago there were numerous lowlands, Khals and channels within and around Chattogram that would drain the City efficiently. The city has spread over the years in all directions and such expansion has caused shrinkage of the natural drainage and wetlands. analyze these changes.

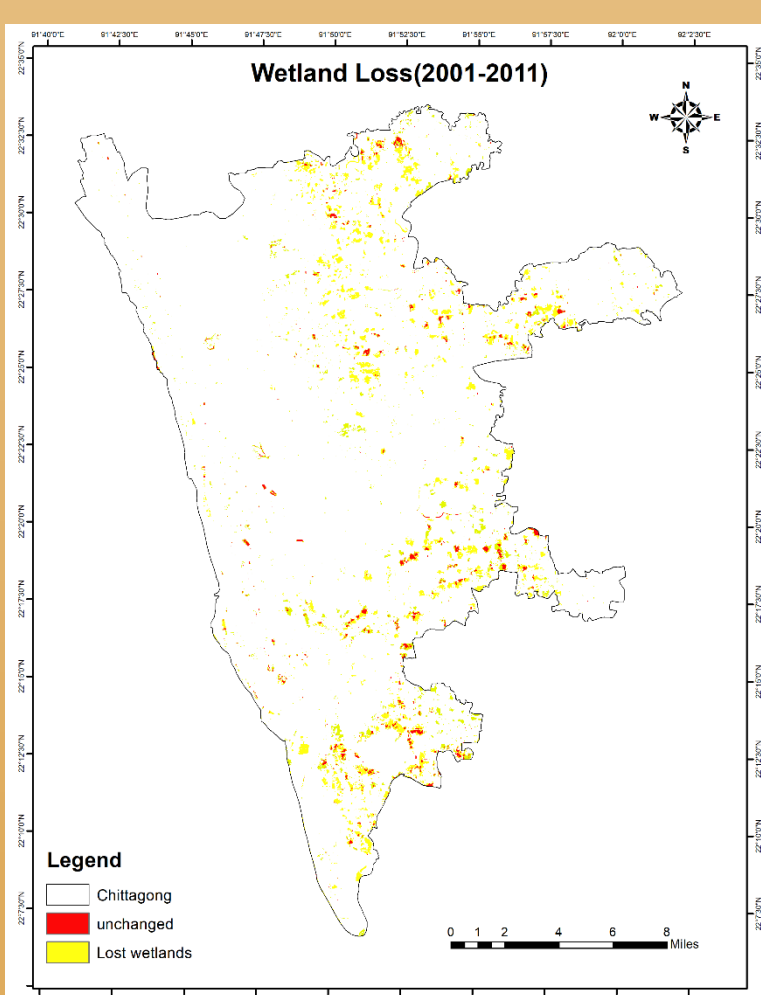
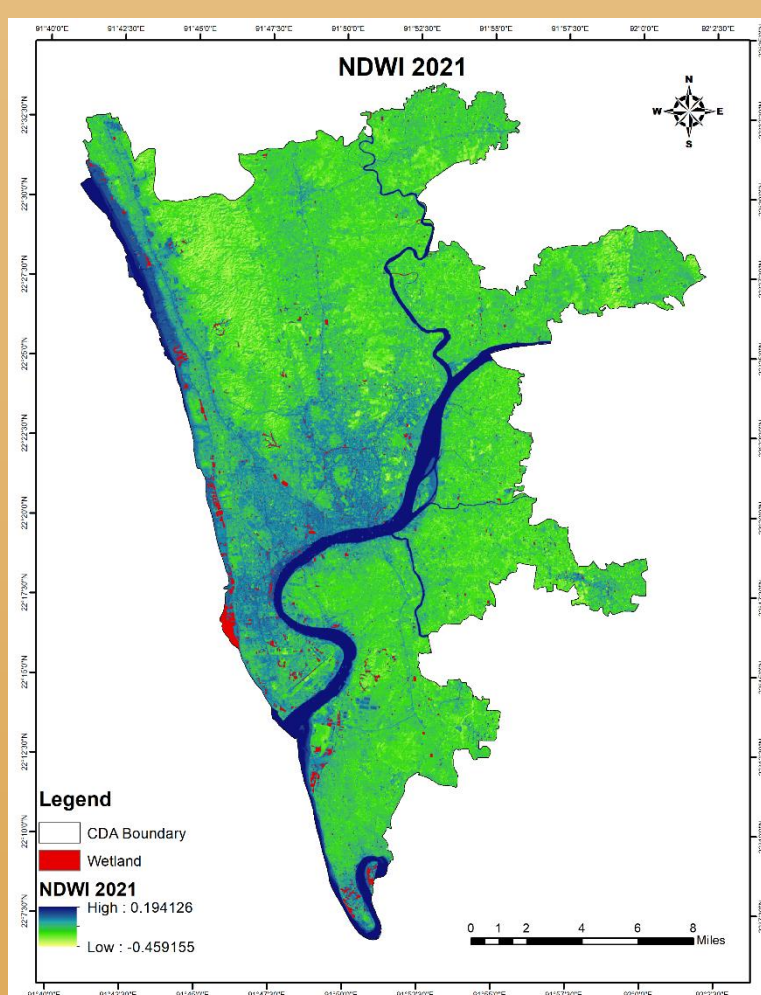
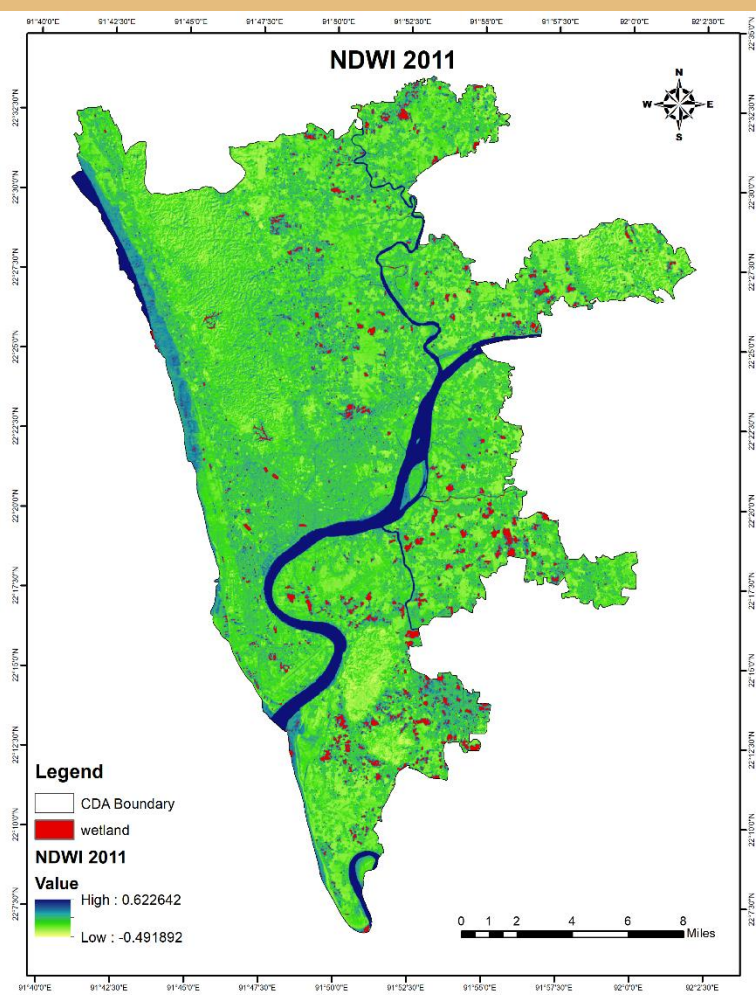
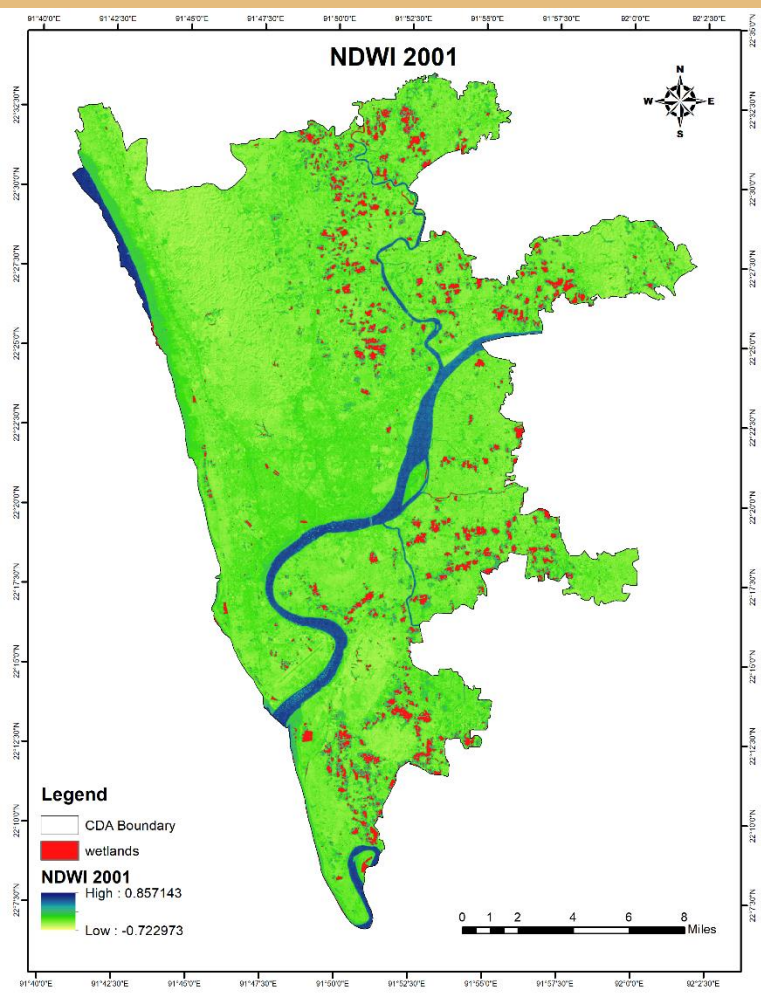
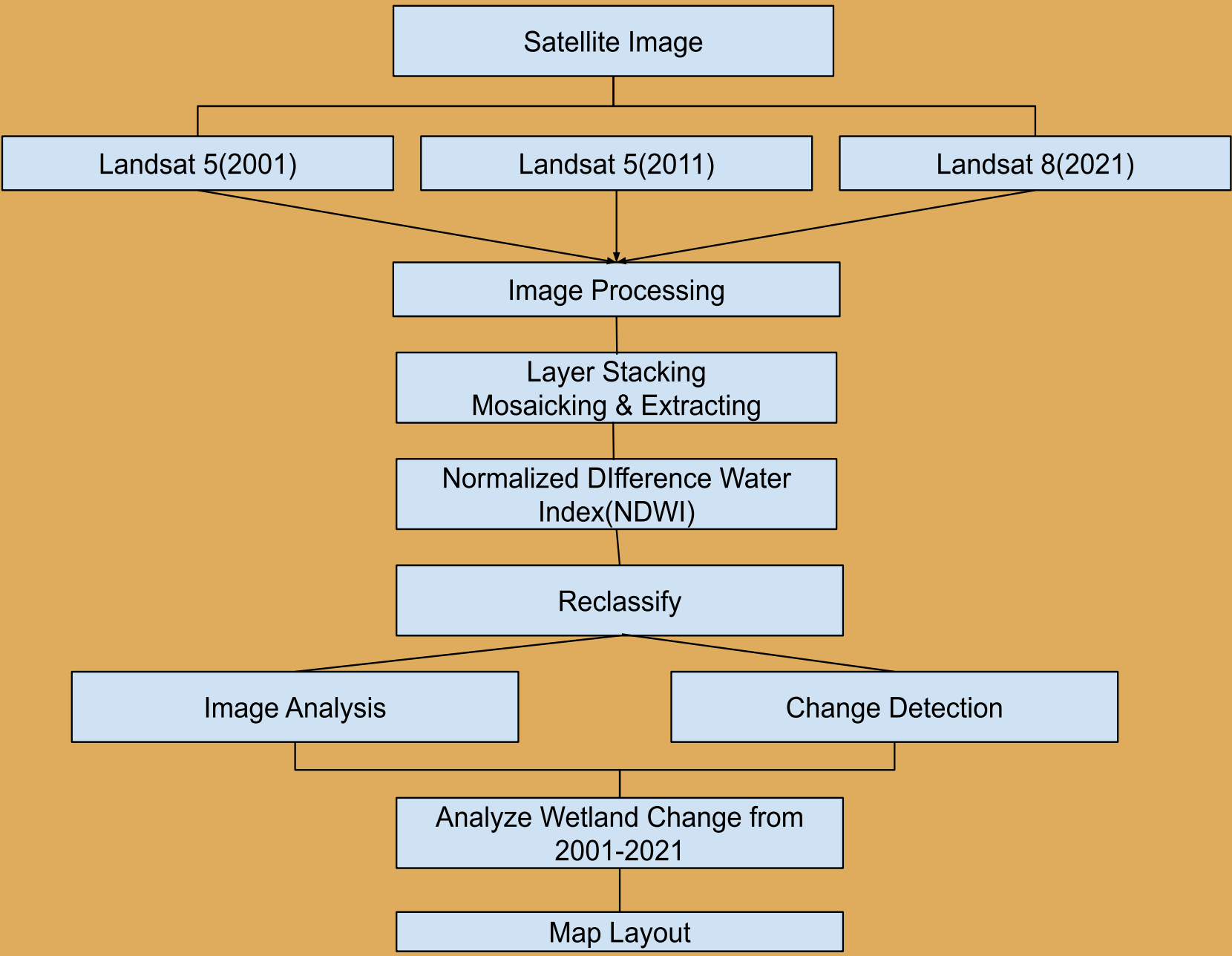
Study Area

Chittagong is the second largest city and prime sea port and the heart of commercial and business activities in Bangladesh. It is lies within 22°14' and 22°24' N Latitude and between 91°46' and 91°53' E Longitude . CDA boundary has an area of 729.85 sq.km. Chittagong city is located towards south-east of the capital city of Dhaka on the bank of Karnaphuli River and surrounded by rich natural resources like green hilly terrain, sea etc.

Goal

The aims of this study therefore were to identify the past and present spatial extent of wetlands, their changing patterns during 2001-2021 of CDA Boundary and make a comparison and analyze these change

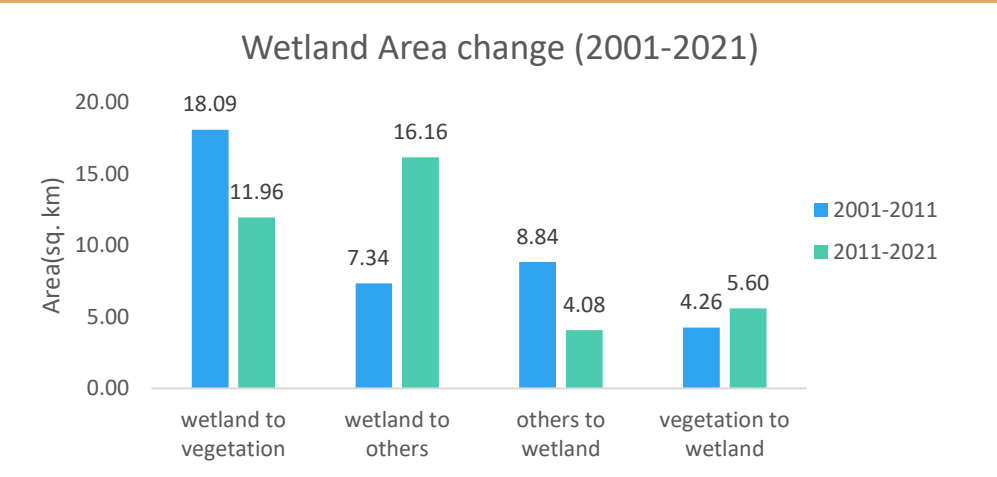
Methodology



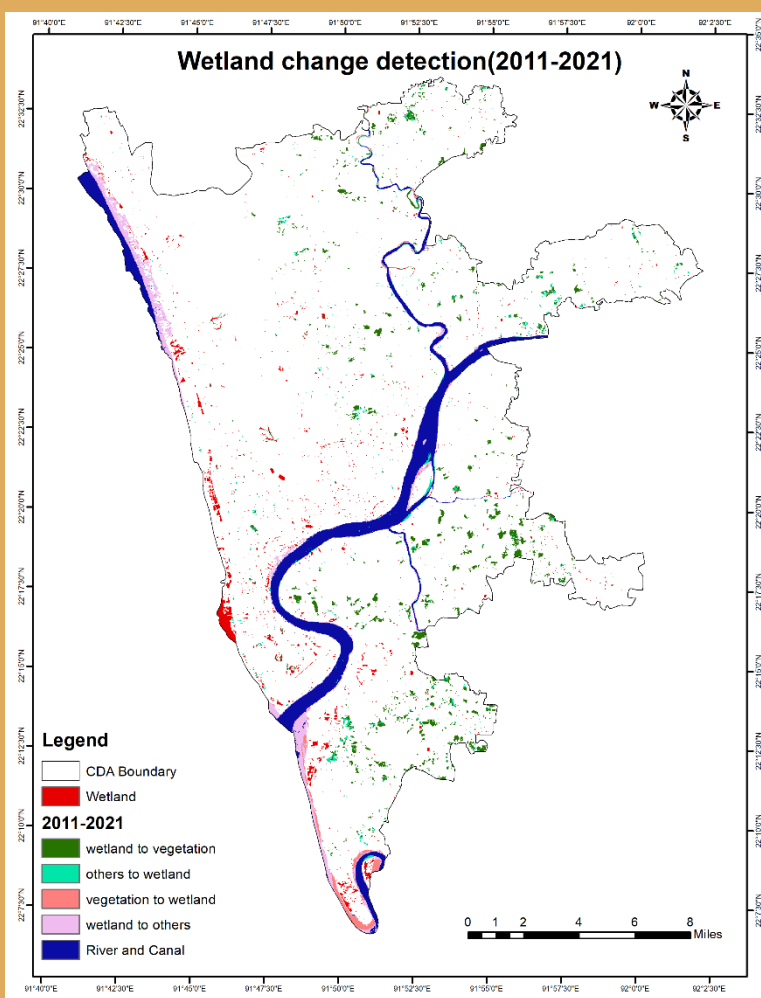
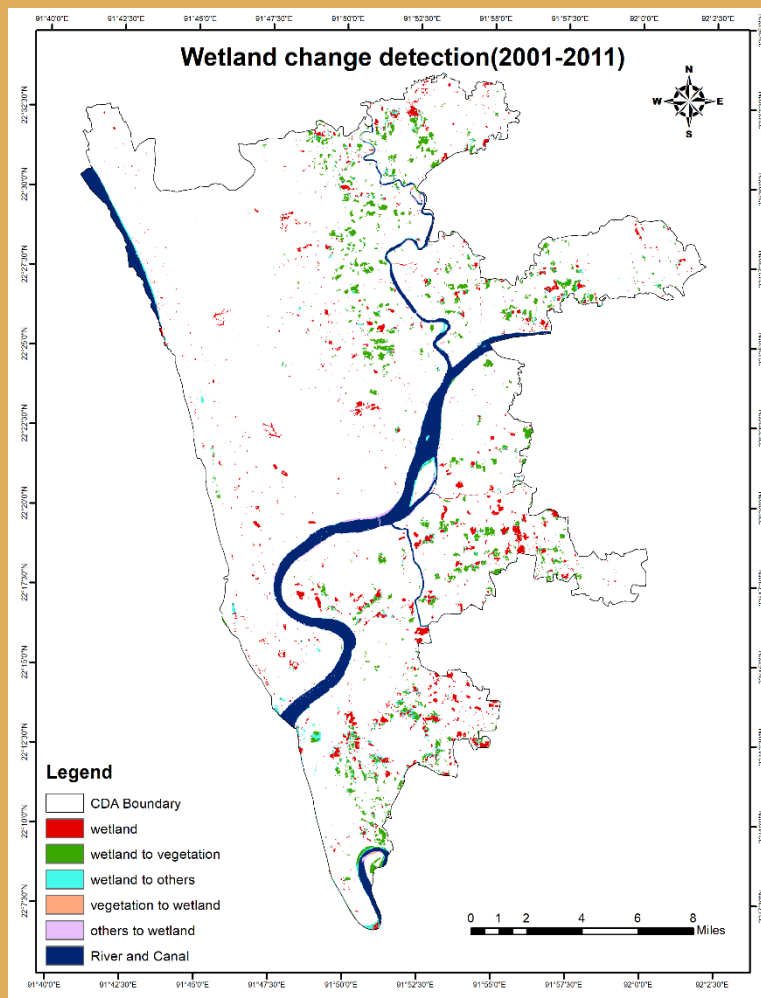
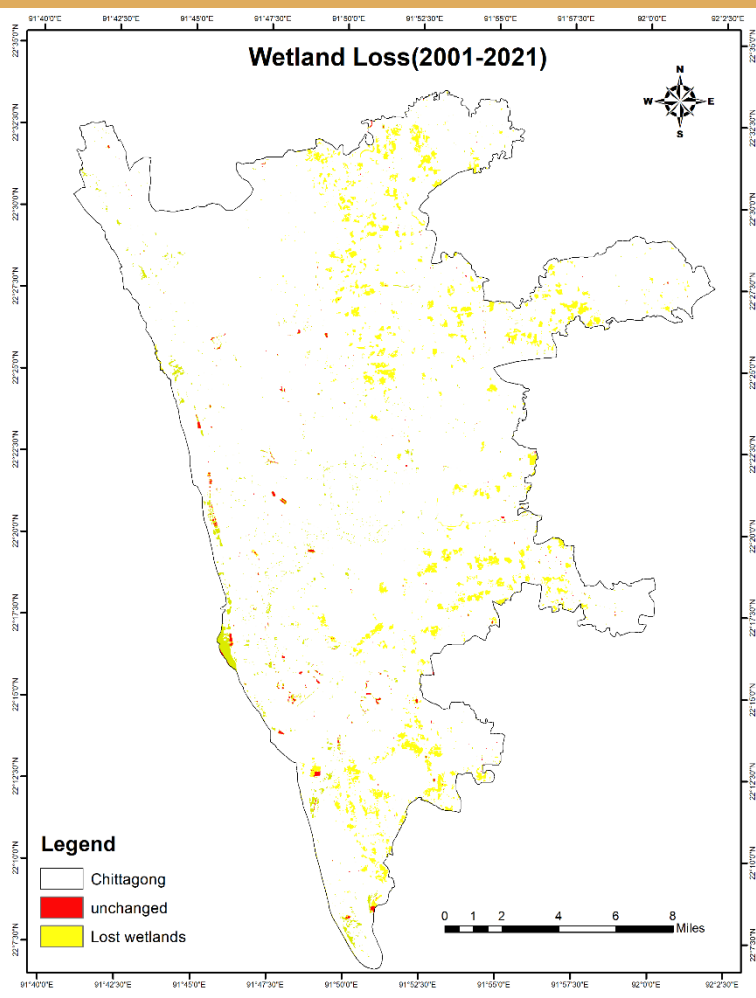
Analysis

Year	Wetland area(in sq km.)	Percentage of Total Area	% of wetland Loss
2001	28.3212	3.880413784	-
2011	16.6563	2.282153867	-41.1878734
2021	8.6265	1.181955196	-48.20878586

2001-2011		
Changed area(sq km.)		
1	wetland to vegetation	18.0855
2	wetland to others	7.3404
3	wetland to others	8.838
4	vegetation to wetland	4.2561
2011-2021		
1	wetland to vegetation	11.9556
2	wetland to others	4.0842
3	wetland to others	16.155
4	vegetation to wetland	5.6025



Over the period 2001 to 2021 total number of wetland converted to other land use is greater than other land use converted to wetland that results a decrease in wetland area of Chattogram city



Data Collection

ID	Satellite Data	DATE	Path	Row
1	Landsat 8	4/07/2021	136	44,45
2	Landsat 5	2/11/2011	136	44,45
3	Landsat 5	01/30/2001	136	44,45

Findings

- There is a significant change in wetland from 2001 to 2021 in CDA boundary
- From 2001-2011 time period reduction in wetland change -41.18% and from 2011-2021 it is -48.20% which shows higher reduction rate in 2011-2021, which means wetlands are decreasing.
- From 2001 to 2011 only 5.3 sq km. of wetlands remained unchanged and in 2011-2021 period only 1.53 sq km. unchanged which shows a lot of wetland converted into other land due to rapid urbanization